

# Harshita Mhaske

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## SUMMARY

Computer Engineering Graduate with hands-on experience in Legged locomotion, Reinforcement learning, and Embedded systems. Skilled in sim-to-real transfer for quadruped robots and passionate about developing intelligent robots that bridge simulation and real-world deployment.

## EDUCATION

**California State University, Fullerton, MS in Computer Engineering** Aug 2023 - May 2025

Relevant Subjects- Artificial Intelligence, Real-time Operating Systems for embedded systems, Artificial Neural Networks, Robot manipulation and planning. **Graduate Thesis** - Legged Robotics under Dr Jidong Huang

## SKILLS

Reinforcement Learning, Machine Learning, Artificial Intelligence, Computer Vision, Manipulation, Path Planning, **Programming** - C++, Python, ROS, Git, Docker, Linux, Gazebo, Pybullet, Tensorflow, OpenCV, PyTorch

## WORK EXPERIENCE

**Graduate Research Assistant** (Sep 2024 – May 2025)

**Institute of Navigation (ION) Lab**, Advisor – Dr. Yu Bai

- Built an open source low-cost (under \$1000) 12 DOF 3D-printed **quadruped robot** capable of walking on uneven terrain
- Integrated **FSRs and IMU** to feed sensory data to linear policy and successfully tested on uneven ground

**Graduate Assistant** (Jan 2025 – May 2025)

**ACCESS Project, CSUF**, Advisor – Dr. Doina Bein

- Mentored underrepresented engineering and computer science undergraduates, enhancing retention by providing academic support and hands-on instruction in C++, Python, machine learning and Robotics

**NASA L'SPACE PROGRAM- NPWEE, Arizona State University** (Mar 2024- Aug 2024)

- Collaborated in the NASA Proposal Writing and Evaluation Experience (NPWEE) under the L'space Program
- Engaged in development and assessment of proposals addressing NASA pain points within Taxonomy 4.1 - Robotics and Sensors

**Robotics Software Engineer** (Feb 2023 – May 2023)

**ARTPARK, Indian Institute of Science, Bangalore**

- Simulated and trained quadruped walking policies using the Motion Imitation framework in PyBullet
- Tested a 1:16 multi-stage planetary gear reduction delivering torque output of 80+Nm

## PROJECTS

**MINIMAL ROBOT DOG - Graduate Thesis**

- Researched the **effect of augmentation of observation space** of a linear policy with newer state variables- ground contacts and force sensors on performance of the policy
- Emulated **contact sensors** using PyBullet environment to train a linear policy

**SCHVAAN ROBOT DOG**

- Built a 12 DOF robot dog with self-built 3D printed **cycloidal actuators of 1:10 gear reduction**
- Tested the accuracy and reliability of actuators for torque, back-drivability and low backlash

**HUMAN ROBOT INTERACTION USING POINTING GESTURES**

- Developed a real-time system using OpenCV that combines MediaPipe **Pose Estimation** with YOLOv4 object detection to identify objects pointed/gestured at by the user
- Implemented geometric calculations to **estimate ground intersection points** based on body landmarks for enhanced spatial interaction

## PUBLICATIONS

- Publication: (ICCAR 2025- Accepted): [Ground contact and reaction force sensing for linear policy control of quadruped robot](#)
- An article about the Schvaan robot was published in a leading National newspaper. – [Link](#)